

SnapSafe AI — Brief Project Description

Tagline: “See it. Detect it. Protect it. Before you share it.” | **Platform Target:** HP Snapdragon AI PCs

1. Project Overview & Problem Statement

Screen sharing during remote interviews, online college classes, code reviews, and customer webinars is a cornerstone of daily productivity. However, it is also one of the leading vectors for accidental data exposure—revealing live API keys in IDE terminals, student IDs, personal phone numbers, passwords, and confidential document headers. Traditional 'cloud AI assistants' paradoxically require streaming unredacted screenshots to external servers, violating corporate and personal privacy. SnapSafe AI solves this as an on-device privacy firewall.

2. The Solution & Technical Innovation

SnapSafe AI intercepts screen frames locally and performs real-time optical character recognition, sensitive pattern detection, and CVSS-weighted risk assessment with a **strict 0-byte cloud egress guarantee**. Key architectural highlights:

- **100% On-Device Inference:** Screen buffers, text tokens, and personal credentials never leave local memory.
- **Qualcomm AI Hub Abstraction:** Features a modular BaseOCRProvider ready for Qualcomm Hexagon NPU execution via QNNExecutionProvider.
- **Entropy & Checksum Rigor:** Distinguishes genuine API keys from natural language via Shannon entropy (> 3.2 bits/symbol) and validates card numbers with Luhn Mod-10.
- **Visual Redaction & Safe Share:** Provides Gaussian blur, pixelation, and blackout with an interactive Before/After split comparison slider.

3. Snapdragon AI PC Optimization

Continuous computer vision on legacy x86 CPUs drains 25W–45W, inducing severe fan noise and battery depletion during video calls. On HP Snapdragon AI PCs, the dedicated 45 TOPS Qualcomm Hexagon NPU executes quantized INT8 vision models at under 5W, ensuring silent, zero-lag performance while leaving the CPU completely free for video conferencing software.

4. Current Implementation Status & Verification

The project is a fully functional MVP with 25 passing automated unit tests, production-compiled React/TypeScript frontend, and FastAPI backend. In the current prototype host, inference runs on local CPU/ONNX fallback, transparently disclosing hardware metrics without fabricated statistics. Full support is architected for immediate QNN context deployment upon hardware transfer.

5. Market Impact, Deployment & Accessibility

SnapSafe AI protects students against academic identity theft, software engineers against catastrophic credential leaks, and healthcare workers against HIPAA violations. The application packages into a lightweight Windows 11 desktop shell with full keyboard accessibility and high-contrast styling.